Homework 5 — Static Magnetic Fields

Complete Solutions with Explanations

This document works through all 22 problems from the Static Magnetic Fields problem set. For each problem I try to show not just the algebra but the physical reasoning — what symmetry tells us, which theorem to reach for, and why the answer has the form it does. Notation: boldface vectors, hats for unit vectors ($\hat{\mathbf{a}}_x$, $\hat{\mathbf{a}}_\phi$, etc.), $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}$, and $\mathbf{B}$, $\mathbf{H}$, $\mathbf{M}$ for magnetic flux density, field intensity, and magnetization respectively.

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Problem 5-1 — Crossed E and B fields (velocity selector)

Setup. A point charge Q moves with velocity $\mathbf{u} = \hat{\mathbf{a}}_x u_0$ through a region where $\mathbf{B} = \hat{\mathbf{a}}_x B_x + \hat{\mathbf{a}}_y B_y + \hat{\mathbf{a}}_z B_z$. We want the electric field $\mathbf{E}$ that keeps the velocity unchanged.

Reasoning. The Lorentz force law reads $\mathbf{F} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$. For no change in velocity, the net force must vanish, so $\mathbf{E} = -\mathbf{u} \times \mathbf{B}$.

$$\begin{aligned}\mathbf{u} \times \mathbf{B} &= (\hat{\mathbf{a}}_x u_0) \times (\hat{\mathbf{a}}_x B_x + \hat{\mathbf{a}}_y B_y + \hat{\mathbf{a}}_z B_z) \\ &= u_0 B_x \underbrace{(\hat{\mathbf{a}}_x \times \hat{\mathbf{a}}_x)}_{=0} + u_0 B_y \underbrace{(\hat{\mathbf{a}}_x \times \hat{\mathbf{a}}_y)}_{=\hat{\mathbf{a}}_z} + u_0 B_z \underbrace{(\hat{\mathbf{a}}_x \times \hat{\mathbf{a}}_z)}_{=-\hat{\mathbf{a}}_y} \\ &= u_0 B_y\, \hat{\mathbf{a}}_z - u_0 B_z\, \hat{\mathbf{a}}_y.\end{aligned}$$

Therefore

$$\boxed{\mathbf{E} = -\mathbf{u} \times \mathbf{B} = \hat{\mathbf{a}}_y\, u_0 B_z - \hat{\mathbf{a}}_z\, u_0 B_y.}$$

Physical note. The x-component of $\mathbf{B}$ does nothing because $\mathbf{u} \parallel \mathbf{B}_x$ — a charge moving along $\mathbf{B}$ feels no magnetic force. Only the perpendicular components deflect the charge, and $\mathbf{E}$ exactly cancels that deflection. This is the principle of a Wien velocity selector: only particles with speed $u_0 = E/B_\perp$ pass through undeflected.

Problem 5-2 — Toroid with rectangular cross section

A toroid with inner radius a, outer radius b, rectangular cross-section of height h, and N closely-wound turns carries current I. We want the exact flux through the cross-section and the percentage error of the "flux-at-mean-radius" approximation.

Step 1 — B inside the toroid

By the azimuthal symmetry of the winding, $\mathbf{B}$ points in $\hat{\mathbf{a}}_\phi$ and depends only on r. Ampère's law on a circular loop of radius r (with $a < r < b$) encloses N turns each carrying I:

$$\oint \mathbf{H} \cdot d\mathbf{l} = H(r) \cdot 2\pi r = NI \quad \Longrightarrow \quad \mathbf{B}(r) = \hat{\mathbf{a}}_\phi \, \frac{\mu_0 NI}{2\pi r}.$$

Step 2 — Exact flux through one cross-section

$$\Phi_{\text{exact}} = \int_a^b B(r)\, h\, dr = \frac{\mu_0 NIh}{2\pi} \int_a^b \frac{dr}{r} = \frac{\mu_0 NIh}{2\pi} \ln\!\left(\frac{b}{a}\right).$$

Step 3 — Approximation using mean radius

With $r_m = (a+b)/2$, the approximation assumes B is uniform across the cross-section at $B(r_m)$:

$$\Phi_{\text{approx}} = B(r_m) \cdot h(b-a) = \frac{\mu_0 NI}{2\pi \cdot (a+b)/2} \cdot h(b-a) = \frac{\mu_0 NIh}{\pi} \cdot \frac{b-a}{a+b}.$$

Step 4 — Error analysis

Let $x = b/a$. Then

$$\frac{\Phi_{\text{exact}}}{\Phi_{\text{approx}}} = \frac{(x+1)\ln x}{2(x-1)}.$$

For $b/a = 5$: $\ln 5 \approx 1.6094$, so

$$\frac{\Phi_{\text{exact}}}{\Phi_{\text{approx}}} = \frac{6 \cdot 1.6094}{2 \cdot 4} = 1.2071.$$

The approximation therefore *under*-estimates the true flux: it gives about 82.8% of the exact value, a relative error of roughly $\boxed{-17.2\%}$.

Intuition. When the toroid is thick (b/a large), $B \propto 1/r$ varies strongly across the cross-section and is much stronger near the inner wall. Evaluating at the mean radius systematically under-weights the intense inner region.

Problem 5-3 — Magnetic field of a finite straight wire

A current I flows along the finite segment P_1P_2. We want $\mathbf{B}$ at a field point P at perpendicular distance r from the wire, with endpoints subtending angles α_1 and α_2 (measured from the perpendicular foot).

Part (a) — Biot–Savart derivation

Place the wire along the z-axis and P in the xy-plane at distance r. A current element $d\mathbf{l}' = \hat{\mathbf{a}}_z\, dz'$ at position $\hat{\mathbf{a}}_z z'$ contributes

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{d\mathbf{l}' \times \hat{\mathbf{R}}}{R^2}.$$

The geometry gives a field in the $\hat{\mathbf{a}}_\phi$ direction with

$$dB_\phi = \frac{\mu_0 I}{4\pi}\,\frac{r\,dz'}{(r^2+z'^2)^{3/2}}.$$

Substitute $z' = r\tan\alpha$, $dz' = r\sec^2\alpha\,d\alpha$, and $(r^2+z'^2)^{3/2} = r^3\sec^3\alpha$:

$$dB_\phi = \frac{\mu_0 I}{4\pi r}\cos\alpha\,d\alpha.$$

Integrating from α_1 to α_2:

$$\boxed{\mathbf{B}_P = \hat{\mathbf{a}}_\phi\,\frac{\mu_0 I}{4\pi r}(\sin\alpha_2 - \sin\alpha_1).}$$

Part (b) — Infinite-wire limit

For an infinite wire, $\alpha_1 \to -\pi/2$ and $\alpha_2 \to +\pi/2$, so $\sin\alpha_2 - \sin\alpha_1 \to 2$, giving

$$\mathbf{B} = \hat{\mathbf{a}}_\phi\,\frac{\mu_0 I}{2\pi r},$$

which is Eq. (5-35). ✓

Problem 5-4 — Field above the current sheet

A sheet of width w lies in the $y=0$ plane (from $x=0$ to $x=w$), infinite in z, carrying current I into the page ($-\hat{\mathbf{a}}_z$ direction). Find $\mathbf{B}$ at $P_1 = (0, d_1, 0)$.

Strategy. Treat the sheet as infinitesimal filaments of width dx' carrying $dI = (I/w)\,dx'$. For a filament at $(x', 0)$, the field at $P_1 = (0, d_1)$ is obtained from the infinite-wire formula.

The vector from the filament to P_1 is $(-x', d_1)$, with distance $R = \sqrt{x'^2 + d_1^2}$. The current element direction is $-\hat{\mathbf{a}}_z$, so $(-\hat{\mathbf{a}}_z)\times\hat{\mathbf{R}}$ gives the field direction. Working it out:

$$d\mathbf{B} = \frac{\mu_0 I}{2\pi w R^2}\,(d_1\,\hat{\mathbf{a}}_x + x'\,\hat{\mathbf{a}}_y)\,dx'.$$

Integrate x' from 0 to w:

$$B_{1x} = \frac{\mu_0 I\,d_1}{2\pi w}\int_0^w \frac{dx'}{x'^2+d_1^2} = \frac{\mu_0 I}{2\pi w}\,\arctan\!\left(\frac{w}{d_1}\right),$$

$$B_{1y} = \frac{\mu_0 I}{2\pi w}\int_0^w \frac{x'\,dx'}{x'^2+d_1^2} = \frac{\mu_0 I}{4\pi w}\,\ln\!\left(1+\frac{w^2}{d_1^2}\right).$$

$$\boxed{\mathbf{B}_1 = \frac{\mu_0 I}{2\pi w}\left[\hat{\mathbf{a}}_x \arctan\frac{w}{d_1} + \frac{\hat{\mathbf{a}}_y}{2}\,\ln\!\left(1+\frac{w^2}{d_1^2}\right)\right].}$$

Sanity check. As $d_1 \to 0$, $\arctan(w/d_1) \to \pi/2$, so $B_{1x} \to \mu_0 I/(4w)$ — exactly half the field of an infinite sheet of surface current density I/w, as expected right at the edge (half the sheet contributes). The y-component diverges logarithmically, as it should right atop a current line.

Problem 5-5 — Field beside the current sheet

Same sheet, but now $P_2 = (w + d_2, 0, 0)$ lies in the plane of the sheet, past the right edge.

The geometry simplifies: for a filament at $(x', 0)$ with current $-\hat{\mathbf{a}}_z$, the point P_2 sits at in-plane distance $(w + d_2) - x'$ along $+x$. By the right-hand rule, the field at P_2 points in $+\hat{\mathbf{a}}_y$:

$$dB_{2y} = \frac{\mu_0\, dI}{2\pi[(w + d_2) - x']} = \frac{\mu_0 I}{2\pi w} \cdot \frac{dx'}{(w + d_2) - x'}.$$

Integrating (substitute $u = (w + d_2) - x'$):

$$\boxed{\mathbf{B}_2 = \hat{\mathbf{a}}_y \frac{\mu_0 I}{2\pi w} \ln\left(1 + \frac{w}{d_2}\right).}$$

Check. For $d_2 \gg w$: $\ln(1 + w/d_2) \approx w/d_2$, giving $\mathbf{B}_2 \approx \hat{\mathbf{a}}_y\, \mu_0 I/(2\pi d_2)$ — the field of a concentrated line current at distance d_2, which is correct for the far-field of a thin sheet.

Problem 5-6 — Coaxial cable

Inner conductor of radius a carries current I in $+\hat{\mathbf{a}}_z$ (uniform density). Outer conductor between b and c carries I in $-\hat{\mathbf{a}}_z$ (also uniform). Field is azimuthal: $\mathbf{B} = \hat{\mathbf{a}}_\phi B_\phi(r)$.

Region 1: $r < a$ (inside inner conductor)

Current density $J_1 = I/(\pi a^2)$; enclosed current $= Ir^2/a^2$:

$$B_\phi(r) = \frac{\mu_0 I r}{2\pi a^2}.$$

Linear rise from 0 to $\mu_0 I/(2\pi a)$ at the surface.

Region 2: $a < r < b$ (insulating gap)

$I_{\text{enc}} = I$:

$$B_\phi(r) = \frac{\mu_0 I}{2\pi r}.$$

Classic $1/r$ falloff.

Region 3: $b < r < c$ (inside outer conductor)

Outer current density $J_3 = -I/[\pi(c^2 - b^2)]$. The enclosed return current within radius r is $-I(r^2 - b^2)/(c^2 - b^2)$, so:

$$I_{\text{enc}} = I \cdot \frac{c^2 - r^2}{c^2 - b^2}, \qquad B_\phi(r) = \frac{\mu_0 I}{2\pi r} \cdot \frac{c^2 - r^2}{c^2 - b^2}.$$

Goes smoothly from $\mu_0 I/(2\pi b)$ at $r = b$ down to 0 at $r = c$.

Region 4: $r > c$

Net enclosed current is zero: $B_\phi = 0$. Coax confines its field perfectly.

Plot of |B| vs r:

$0 \le r \le a$:	linear rise, $0 \to \mu_0 I/(2\pi a)$
$a \le r \le b$:	$1/r$ decay, $\mu_0 I/(2\pi a) \to \mu_0 I/(2\pi b)$
$b \le r \le c$:	smooth drop to zero
$r \ge c$:	identically zero

Problem 5-7 — Field at the center of an equilateral triangle

A wire of total length $3w$ forms an equilateral triangle (side w) carrying current I. Find $\mathbf{B}$ at the center.

Use the finite-wire formula. The perpendicular distance from the centroid to each side (the apothem) is $r = w/(2\sqrt{3})$. From the centroid, each endpoint of a side sits at half-side-length $w/2$ from the perpendicular foot, so

$$\tan\alpha = \frac{w/2}{w/(2\sqrt{3})} = \sqrt{3} \quad \Longrightarrow \quad \alpha = 60^\circ.$$

With $\alpha_2 = +60^\circ$, $\alpha_1 = -60^\circ$: $\sin\alpha_2 - \sin\alpha_1 = \sqrt{3}$. Each side contributes

$$B_{\text{side}} = \frac{\mu_0 I}{4\pi r}\sqrt{3} = \frac{\mu_0 I\sqrt{3}\cdot 2\sqrt{3}}{4\pi w} = \frac{3\mu_0 I}{2\pi w}.$$

All three sides contribute in the same direction (perpendicular to the triangle's plane, by the right-hand rule), so

$$\boxed{\mathbf{B}_{\text{center}} = \hat{\mathbf{a}}_n \frac{9\mu_0 I}{2\pi w}.}$$

Problem 5-8 — Field on the axis of a finite solenoid

Solenoid of radius b, length L, N turns, current I. Find $\mathbf{B}$ on the axis and take $L \to \infty$.

A ring of width dz' at position z' carries $dI = (NI/L)\,dz'$. The on-axis field of a single loop (Eq. 5-37) is

$$dB_z = \frac{\mu_0\,(NI/L)\,b^2\,dz'}{2(b^2 + z'^2)^{3/2}}.$$

Placing the field point P at $z = 0$ and the solenoid from $z' = z_1$ to $z' = z_2$ (signed distances from P):

$$B_z = \frac{\mu_0 N I b^2}{2L}\int_{z_1}^{z_2} \frac{dz'}{(b^2 + z'^2)^{3/2}}.$$

The antiderivative is $z'/[b^2\sqrt{b^2 + z'^2}]$, giving

$$\boxed{B_z = \frac{\mu_0 N I}{2L}\left[\frac{z_2}{\sqrt{b^2 + z_2^2}} - \frac{z_1}{\sqrt{b^2 + z_1^2}}\right] = \frac{\mu_0 n I}{2}(\cos\theta_2 - \cos\theta_1),}$$

where $n = N/L$ and θ_i is the angle from the axis to the rim of each solenoid end.

Infinite-solenoid limit

As $L \to \infty$ with P interior, $z_1 \to -\infty$ and $z_2 \to +\infty$, so the bracketed expression goes to $1 - (-1) = 2$:

$$B_z \to \mu_0 n I,$$

recovering Eq. (5-82). ✓

Problem 5-9 — Vector potential of a thick wire

An infinite wire of radius $R = 2\,\text{mm}$ along the z-axis carries current I. For $r > R$, $\mathbf{B} = \hat{\mathbf{a}}_\phi\, \mu_0 I/(2\pi r)$.

Part (a) — A for $r > R$

By symmetry $\mathbf{A} = \hat{\mathbf{a}}_z A_z(r)$. Since $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}$ gives $B_\phi = -\partial A_z/\partial r$:

$$\frac{\partial A_z}{\partial r} = -\frac{\mu_0 I}{2\pi r} \quad \Longrightarrow \quad A_z(r) = -\frac{\mu_0 I}{2\pi} \ln r + C.$$

Choose $A_z = 0$ at $r = R$:

$$\boxed{\mathbf{A} = \hat{\mathbf{a}}_z \frac{\mu_0 I}{2\pi} \ln\!\left(\frac{R}{r}\right), \qquad r \geq R.}$$

(Negative for $r > R$, since $\ln(R/r) < 0$ there.)

Part (b) — Flux through the square loop

The loop is in the yz-plane ($x = 0$): $z = \pm 0.3\,\text{m}$, y from $0.1\,\text{m}$ to $0.7\,\text{m}$. In this plane, $r = |y|$.

Use the line-integral form of the flux: $\Phi = \oint \mathbf{A} \cdot d\mathbf{l}$. Since $\mathbf{A}$ has only a z-component, only the two sides parallel to z contribute. They lie at $y = 0.1$ and $y = 0.7$, each of length $0.6\,\text{m}$, and are traversed in opposite z-directions:

$$\Phi = 0.6\,[A_z(0.1) - A_z(0.7)] = 0.6 \cdot \frac{\mu_0 I}{2\pi} \ln\!\left(\frac{0.7}{0.1}\right) = 0.6 \cdot \frac{\mu_0 I}{2\pi} \ln 7.$$

With $I = 10\,\text{A}$, $\mu_0 I/(2\pi) = 2 \times 10^{-6}$:

$$\Phi = 0.6 \cdot 2 \times 10^{-6} \cdot \ln 7 \approx 1.2 \times 10^{-6} \cdot 1.9459 \approx \boxed{2.34\,\mu\text{Wb}.}$$

Cross-check by direct integration: $\Phi = \int_{-0.3}^{0.3} dz \int_{0.1}^{0.7} \frac{\mu_0 I}{2\pi y}\, dy = 0.6 \cdot (\mu_0 I/2\pi) \ln 7$. ✓

Problem 5-10 — Infinite current sheet

Surface current density $\mathbf{J}_s = \hat{\mathbf{a}}_x J_{s0}$ in the xy-plane.

Part (a) — B above and below

By symmetry $\mathbf{B} \parallel \hat{\mathbf{a}}_y$, uniform on each side, flipping sign across the sheet. Ampère's law around a rectangular loop straddling the sheet (perpendicular segments contribute zero) gives $2B\ell = \mu_0 J_{s0} \ell$:

$$\boxed{\mathbf{B}(z > 0) = -\hat{\mathbf{a}}_y \frac{\mu_0 J_{s0}}{2}, \qquad \mathbf{B}(z < 0) = +\hat{\mathbf{a}}_y \frac{\mu_0 J_{s0}}{2}.}$$

Part (b) — A from B

Try $\mathbf{A} = \hat{\mathbf{a}}_x f(z)$. Then $\boldsymbol{\nabla} \times \mathbf{A} = -\hat{\mathbf{a}}_y \, \partial f/\partial z$ (only z-dependence). Matching to $\mathbf{B}$:

$$-\frac{\partial f}{\partial z} = \mp \frac{\mu_0 J_{s0}}{2} \quad \text{for} \quad z \gtrless 0.$$

So $f(z) = -(\mu_0 J_{s0}/2)|z| + C$. Choosing $\mathbf{A}(z_0) = 0$:

$$\boxed{\mathbf{A} = \hat{\mathbf{a}}_x \frac{\mu_0 J_{s0}}{2} (|z_0| - |z|).}$$

Problem 5-11 — Slab in an external field

Slab of thickness d perpendicular to $\mathbf{H}_0 = \hat{\mathbf{a}}_z H_0$.

Part (a) — Linear material with permeability μ

Boundary conditions: B_n continuous (no free surface current implies H_t continuous too). Here $\mathbf{B}$ is entirely normal to the faces, so B_z must be continuous. Outside: $B = \mu_0 H_0$. Inside: $B = \mu H_{\text{in}}$. Therefore

$$\boxed{\mathbf{H}_{\text{inside}} = \hat{\mathbf{a}}_z \frac{H_0}{\mu_r}, \qquad \mathbf{B}_{\text{inside}} = \hat{\mathbf{a}}_z \, \mu_0 H_0.}$$

For highly permeable material (μ_r large), $\mathbf{H}$ inside is small — the material "screens" the H-field for this perpendicular geometry — while $\mathbf{B}$ is unchanged.

Part (b) — Permanent magnet, $\mathbf{M}_i = \hat{\mathbf{a}}_z M_i$

Now $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$. Outside, $\mathbf{M} = 0$: $B = \mu_0 H_0$. Inside, $B = \mu_0(H_{\text{in}} + M_i)$. Continuity of B_z:

$$\mu_0(H_{\text{in}} + M_i) = \mu_0 H_0 \quad \Longrightarrow \quad \boxed{\mathbf{H}_{\text{in}} = \hat{\mathbf{a}}_z(H_0 - M_i), \qquad \mathbf{B}_{\text{in}} = \hat{\mathbf{a}}_z \, \mu_0 H_0.}$$

If $M_i > H_0$, $\mathbf{H}_{\text{in}}$ points *opposite* to $\mathbf{H}_0$ — this is the demagnetizing field created by the bound surface currents on the magnet's faces.

Problem 5-12 — Magnetic rod in a solenoid

Rod of radius a, permeability μ, sits coaxially in a long air-cored solenoid of inner radius $b > a$ and turn density n, carrying current I.

Part (a) — B, H, M inside

In the long-solenoid limit $\mathbf{H}$ is determined by the free currents only: $\mathbf{H} = \hat{\mathbf{a}}_z \, nI$ everywhere inside, regardless of material.

For $r < a$ (inside the rod):

$$\mathbf{H} = \hat{\mathbf{a}}_z nI, \quad \mathbf{B} = \hat{\mathbf{a}}_z \, \mu nI, \quad \mathbf{M} = \hat{\mathbf{a}}_z(\mu_r - 1)nI = \hat{\mathbf{a}}_z \, \chi_m nI.$$

For $a < r < b$ (air gap):

$$\mathbf{H} = \hat{\mathbf{a}}_z nI, \quad \mathbf{B} = \hat{\mathbf{a}}_z \, \mu_0 nI, \quad \mathbf{M} = 0.$$

Part (b) — Equivalent magnetization currents

Volume: $\mathbf{J}_{mv} = \boldsymbol{\nabla} \times \mathbf{M} = 0$ (uniform $\mathbf{M}$).

Surface: On the lateral surface $r = a$, outward normal $\hat{\mathbf{a}}_r$:

$$\mathbf{J}_{ms} = \mathbf{M} \times \hat{\mathbf{a}}_n = \hat{\mathbf{a}}_z(\mu_r - 1)nI \times \hat{\mathbf{a}}_r = \hat{\mathbf{a}}_\phi(\mu_r - 1)nI.$$

Physically, the rod behaves like an extra solenoid wound on its surface with effective turn density $(\mu_r - 1)n$, boosting $\mathbf{B}$ inside to $\mu_0\mu_r nI$.

Problem 5-13 — Uniformly magnetized sphere

Sphere of radius b with $\mathbf{M} = \hat{\mathbf{a}}_z M_0$.

Part (a) — Bound currents

Volume: $\mathbf{J}_{mv} = \boldsymbol{\nabla} \times \mathbf{M} = 0$.

Surface: $\mathbf{J}_{ms} = \mathbf{M} \times \hat{\mathbf{a}}_n = M_0 \hat{\mathbf{a}}_z \times \hat{\mathbf{a}}_r$. Using $\hat{\mathbf{a}}_z = \cos\theta\,\hat{\mathbf{a}}_r - \sin\theta\,\hat{\mathbf{a}}_\theta$ and $\hat{\mathbf{a}}_\theta \times \hat{\mathbf{a}}_r = -\hat{\mathbf{a}}_\phi$:

$$\boxed{\mathbf{J}_{ms} = M_0 \sin\theta\,\hat{\mathbf{a}}_\phi.}$$

The bound current circulates azimuthally, maximal at the equator and vanishing at the poles.

Part (b) — B at the center

A classical result: inside a uniformly magnetized sphere the field is uniform and equal to $\frac{2}{3}\mu_0\mathbf{M}$. Therefore

$$\boxed{\mathbf{B}(0) = \hat{\mathbf{a}}_z\,\tfrac{2}{3}\mu_0 M_0.}$$

Why 2/3? The $\sin\theta$ surface current on a sphere is equivalent to the current distribution of two uniformly-charged spheres rotating about the same axis but slightly offset (the Poisson picture), which produces a uniform interior field of magnitude $(2/3)\mu_0 M$.

Problem 5-14 — Air–iron interface

Plane $y = 0$: air (1, $\mu_{r1} = 1$) and iron (2, $\mu_{r2} = 5000$). Normal direction is $\hat{\mathbf{a}}_y$; tangential directions are $\hat{\mathbf{a}}_x$ and $\hat{\mathbf{a}}_z$.

Boundary conditions (no free current): B_n continuous, H_t continuous, i.e. $B_{1t}/\mu_1 = B_{2t}/\mu_2$.

Part (a) — Given $\mathbf{B}_1 = \hat{\mathbf{a}}_x 2 - \hat{\mathbf{a}}_y 10$ (mT)

$$\begin{aligned}
\mathbf{B}_{1t} &= \hat{\mathbf{a}}_x 2 \text{ mT}, \quad \mathbf{B}_{1n} = -\hat{\mathbf{a}}_y 10 \text{ mT}, \\
\mathbf{B}_{2n} &= \mathbf{B}_{1n} = -\hat{\mathbf{a}}_y 10 \text{ mT}, \\
\mathbf{B}_{2t} &= \frac{\mu_2}{\mu_1}\mathbf{B}_{1t} = 5000 \cdot \hat{\mathbf{a}}_x 2 \text{ mT} = \hat{\mathbf{a}}_x\, 10 \text{ T}.
\end{aligned}$$

$$\boxed{\mathbf{B}_2 = \hat{\mathbf{a}}_x\, 10 \text{ T} - \hat{\mathbf{a}}_y\, 10 \text{ mT}.}$$

Angle with the interface: $\tan\alpha = |B_n|/|B_t| = 0.010/10 = 10^{-3}$, so $\alpha \approx 0.057°$ — the field lies almost perfectly parallel to the interface in the iron, as expected for highly permeable material.

Part (b) — Given $\mathbf{B}_2 = \hat{\mathbf{a}}_x 10 + \hat{\mathbf{a}}_y 2$ (mT)

$$\mathbf{B}_{2t} = \hat{\mathbf{a}}_x 10 \text{ mT}, \quad \mathbf{B}_{2n} = \hat{\mathbf{a}}_y 2 \text{ mT},$$
$$\mathbf{B}_{1n} = \mathbf{B}_{2n} = \hat{\mathbf{a}}_y 2 \text{ mT},$$
$$\mathbf{B}_{1t} = \frac{\mu_1}{\mu_2}\mathbf{B}_{2t} = \tfrac{1}{5000} \cdot \hat{\mathbf{a}}_x 10 \text{ mT} = \hat{\mathbf{a}}_x\, 2\,\mu\text{T}.$$

$$\boxed{\mathbf{B}_1 = \hat{\mathbf{a}}_x 2\,\mu\text{T} + \hat{\mathbf{a}}_y 2 \text{ mT}.}$$

Angle from the normal: $\tan\alpha = |B_t|/|B_n| = 10^{-3}$, so $\alpha \approx 0.057°$ — air-side field is nearly normal to the interface.

Problem 5-15 — Self-inductance of a toroid (circular cross-section)

Toroid of mean radius r_0, circular cross-section radius b, N turns.

Field inside

Same Amperian-loop argument: $\mathbf{B}(r) = \hat{\mathbf{a}}_\phi\, \mu_0 NI/(2\pi r)$.

Flux through one turn

Parametrize the circular cross-section by polar coordinates (ρ, φ) centered on the cross-section's center, where φ is measured from the direction toward the toroid axis, so $r = r_0 - \rho\cos\varphi$. With $dA = \rho\, d\rho\, d\varphi$:

$$\Phi_1 = \int_0^{2\pi}\int_0^{b} \frac{\mu_0 NI}{2\pi(r_0 - \rho\cos\varphi)}\,\rho\, d\rho\, d\varphi.$$

Use $\int_0^{2\pi} d\varphi/(A - B\cos\varphi) = 2\pi/\sqrt{A^2 - B^2}$ for $A > B$:

$$\Phi_1 = \frac{\mu_0 NI}{2\pi}\cdot 2\pi \int_0^{b} \frac{\rho\, d\rho}{\sqrt{r_0^2 - \rho^2}} = \mu_0 NI\left[r_0 - \sqrt{r_0^2 - b^2}\right].$$

Inductance

All N turns link Φ_1, so $\lambda = N\Phi_1$ and $L = \lambda/I$:

$$\boxed{L = \mu_0 N^2\left[r_0 - \sqrt{r_0^2 - b^2}\right]. \text{ (exact)}}$$

Approximation for $b \ll r_0$

Expand $\sqrt{r_0^2 - b^2} \approx r_0 - b^2/(2r_0)$:

$$\boxed{L \approx \frac{\mu_0 N^2 b^2}{2r_0} = \frac{\mu_0 N^2 \pi b^2}{2\pi r_0} = \frac{\mu_0 N^2 A}{\ell},}$$

which is the classic solenoid formula with cross-sectional area $A = \pi b^2$ and winding length $\ell = 2\pi r_0$.

Problem 5-16 — Mutual inductance: wire and triangular loop

Long wire carries I_1. Coplanar equilateral triangle of side b has one vertex pointing at the wire, nearest vertex at perpendicular distance d, opposite side parallel to the wire.

At distance r from the wire, $B = \mu_0 I_1/(2\pi r)$. The triangle extends from $r = d$ (vertex) to $r = d + b\sqrt{3}/2$ (height of an equilateral triangle is $b\sqrt{3}/2$). At position r, the triangle's width (along the direction parallel to the wire) grows linearly from 0 to b:

$$\text{width}(r) = \frac{b}{b\sqrt{3}/2}\,(r-d) = \frac{2}{\sqrt{3}}(r-d).$$

Flux:

$$\begin{aligned}\Phi &= \int_d^{d+b\sqrt{3}/2} \frac{\mu_0 I_1}{2\pi r}\cdot\frac{2(r-d)}{\sqrt{3}}\,dr \\ &= \frac{\mu_0 I_1}{\pi\sqrt{3}}\int_d^{d+b\sqrt{3}/2}\left(1-\frac{d}{r}\right)dr \\ &= \frac{\mu_0 I_1}{\pi\sqrt{3}}\left[\frac{b\sqrt{3}}{2} - d\ln\left(1+\frac{b\sqrt{3}}{2d}\right)\right].\end{aligned}$$

$$\boxed{M = \frac{\Phi}{I_1} = \frac{\mu_0 b}{2\pi} - \frac{\mu_0 d}{\pi\sqrt{3}}\ln\left(1+\frac{b\sqrt{3}}{2d}\right).}$$

Problem 5-17 — Mutual inductance: two rectangles

Coplanar rectangles: big one ($w_1 \times h_1$), small one ($w_2 \times h_2$), separation d between facing edges. Assume $h_1 \gg h_2$, $h_2 > w_2 > d$.

Because $h_1 \gg h_2$, the two long vertical sides of the big loop dominate; treat them as infinite wires carrying I_1 (in opposite directions, since they're the two sides of a loop).

Near wire contribution (distance d to $d + w_2$, field over height h_2):

$$\Phi_{\text{near}} = \frac{\mu_0 I_1 h_2}{2\pi}\ln\left(\frac{d+w_2}{d}\right).$$

Far wire contribution (distance $w_1 - d - w_2$ to $w_1 - d$), same direction because opposite-current-at-farther-distance adds to flux:

$$\Phi_{\text{far}} = \frac{\mu_0 I_1 h_2}{2\pi}\ln\left(\frac{w_1-d}{w_1-d-w_2}\right).$$

$$\boxed{M = \frac{\mu_0 h_2}{2\pi}\left[\ln\left(\frac{d+w_2}{d}\right) + \ln\left(\frac{w_1-d}{w_1-d-w_2}\right)\right].}$$

To leading order when w_1 is much larger than the other lengths, the second term is negligible and

$$M \approx \frac{\mu_0 h_2}{2\pi}\ln\left(1+\frac{w_2}{d}\right).$$

Problem 5-18 — Three parallel wires

Three infinite parallel wires at the vertices of an equilateral triangle of side $s = 10\,\text{cm}$, each carrying $I = 25\,\text{A}$ in the same direction.

Consider the top wire. Each neighbor is at distance s and exerts a pairwise attractive force per unit length

$$f_1 = \frac{\mu_0 I^2}{2\pi s} = \frac{(4\pi \times 10^{-7})(25)^2}{2\pi(0.10)} = 1.25 \times 10^{-3}\ \text{N/m}.$$

By symmetry the two neighbor-forces are mirror images about the vertical; their horizontal components cancel. Each makes angle 30° with the downward vertical, so the vertical components add:

$$f_{\text{total}} = 2 f_1 \cos 30^\circ = \sqrt{3} \cdot 1.25 \times 10^{-3} \approx \boxed{2.17 \times 10^{-3}\ \text{N/m}.}$$

Direction: toward the centroid of the triangle. By symmetry, each of the three wires experiences the same magnitude, each pointing inward to the centroid.

Problem 5-19 — Strip and parallel wire, opposite currents

Strip of width w in the plane $x = 0$, parallel wire at distance D (perpendicular to the strip's plane), equal and opposite currents.

Treat the strip as filaments at positions $(0, y')$, each carrying current $(I/w)\,dy'$, with $y' \in [-w/2, w/2]$. The pairwise force between a strip filament and the wire (opposite currents, repulsive) separated by $r = \sqrt{D^2 + y'^2}$ has magnitude $\mu_0 (I/w)(I)/(2\pi r)$ per unit length.

By symmetry only the x-component (perpendicular to the strip's plane) survives the integration. That component is $\cos\theta = D/r$ of the pairwise force:

$$\frac{dF_x}{L} = -\frac{\mu_0 I^2}{2\pi w} \cdot \frac{D\,dy'}{D^2 + y'^2}.$$

Integrate from $-w/2$ to $+w/2$:

$$\frac{F_x}{L} = -\frac{\mu_0 I^2 D}{2\pi w} \cdot \frac{2}{D} \arctan\frac{w}{2D} = -\frac{\mu_0 I^2}{\pi w} \arctan\frac{w}{2D}.$$

Magnitude:

$$\boxed{\left|\frac{F}{L}\right| = \frac{\mu_0 I^2}{\pi w} \arctan\frac{w}{2D} \quad \text{(repulsive, opposite currents).}}$$

Check. As $w \to 0$ with D fixed, $\arctan(w/2D) \approx w/2D$, and the formula reduces to $\mu_0 I^2/(2\pi D)$ — the standard two-wire result. ✓

Problem 5-20 — Force on the connecting bar

Two long parallel lines of radius b, separated by distance d, with currents I in opposite directions, connected at one end by a transverse bar AA' also carrying I.

Place wire 1 at $(0, 0)$ (current $+\hat{\mathbf{a}}_z$) and wire 2 at $(d, 0)$ (current $-\hat{\mathbf{a}}_z$). The bar runs along x from $x = b$ to $x = d - b$, carrying current in $+\hat{\mathbf{a}}_x$.

At position x on the bar:

$$\mathbf{B}(x) = \hat{\mathbf{a}}_y \frac{\mu_0 I}{2\pi}\left[\frac{1}{x} + \frac{1}{d-x}\right].$$

(Both contributions add because the current in wire 2 is reversed but sits on the other side.)

Force per unit length on the bar: $d\mathbf{F}/dl = I\,\hat{\mathbf{a}}_x \times \mathbf{B}$, with $\hat{\mathbf{a}}_x \times \hat{\mathbf{a}}_y = \hat{\mathbf{a}}_z$. Total force:

$$\begin{aligned} F_z &= I\int_b^{d-b} \frac{\mu_0 I}{2\pi}\left[\frac{1}{x} + \frac{1}{d-x}\right] dx \\ &= \frac{\mu_0 I^2}{2\pi}\left[\ln x - \ln(d-x)\right]_b^{d-b} \\ &= \frac{\mu_0 I^2}{2\pi}\left[2\ln(d-b) - 2\ln b\right] \\ &= \frac{\mu_0 I^2}{\pi}\,\ln\!\left(\frac{d-b}{b}\right). \end{aligned}$$

$$\boxed{\mathbf{F} = \hat{\mathbf{a}}_z \frac{\mu_0 I^2}{\pi}\,\ln\!\left(\frac{d-b}{b}\right).}$$

Direction: along the parallel lines, away from the connected end. This is the "blast force" that tries to blow apart the contacts of a circuit-breaker under high current — a real safety concern in power engineering.

Problem 5-21 — Forces and torque on a triangular loop

Equilateral-like triangle with vertices $(-10, 0)$, $(10, 0)$, $(0, 20)$ (all in cm), carrying $I = 10\,\text{A}$ counterclockwise, in uniform $\mathbf{B} = \hat{\mathbf{a}}_y\, 6\,\text{mT}$.

Forces on each side

Side 1 (bottom): $\mathbf{L}_1 = \hat{\mathbf{a}}_x\, 0.20\,\text{m}$.

$$\mathbf{F}_1 = I\mathbf{L}_1 \times \mathbf{B} = 10\,(0.20\hat{\mathbf{a}}_x) \times (0.006\hat{\mathbf{a}}_y) = \hat{\mathbf{a}}_z\,(0.012)\ \text{N} = \hat{\mathbf{a}}_z\, 12\ \text{mN}.$$

Side 2 (right): $\mathbf{L}_2 = (-0.10\hat{\mathbf{a}}_x + 0.20\hat{\mathbf{a}}_y)\,\text{m}$.

$$\mathbf{F}_2 = 10\,(-0.10\hat{\mathbf{a}}_x + 0.20\hat{\mathbf{a}}_y) \times (0.006\hat{\mathbf{a}}_y) = 10\,(-0.10)(0.006)\,\hat{\mathbf{a}}_z = -\hat{\mathbf{a}}_z\, 6\ \text{mN}.$$

Side 3 (left): $\mathbf{L}_3 = (-0.10\hat{\mathbf{a}}_x - 0.20\hat{\mathbf{a}}_y)\,\text{m}$.

$$\mathbf{F}_3 = 10\,(-0.10\hat{\mathbf{a}}_x - 0.20\hat{\mathbf{a}}_y) \times (0.006\hat{\mathbf{a}}_y) = -\hat{\mathbf{a}}_z\, 6\ \text{mN}.$$

Check: $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = (12 - 6 - 6)\hat{\mathbf{a}}_z = 0$. ✓

Torque

Area: $A = \frac{1}{2}(0.20)(0.20) = 0.020\,\text{m}^2$. Magnetic moment:

$$\mathbf{m} = IA\,\hat{\mathbf{a}}_n = (10)(0.020)\,\hat{\mathbf{a}}_z = 0.20\,\hat{\mathbf{a}}_z\ \text{A}\cdot\text{m}^2.$$

$$\boxed{\tau = \mathbf{m} \times \mathbf{B} = (0.20\hat{\mathbf{a}}_z) \times (0.006\hat{\mathbf{a}}_y) = -\hat{\mathbf{a}}_x\, 1.2 \times 10^{-3}\ \text{N}\cdot\text{m}.}$$

The torque tilts the loop about the x-axis so that $\mathbf{m}$ (currently along $+\hat{\mathbf{a}}_z$) rotates toward $\mathbf{B}$ (along $+\hat{\mathbf{a}}_y$) — the minimum-energy alignment.

Problem 5-22 — Torque on a small loop inside a large loop

Small loop (radius r_1, current I_1) at the center of a large loop (radius $r_2 \gg r_1$, current I_2, same sense). Angle between normals is θ.

Since $r_2 \gg r_1$, the field at the small loop is essentially the uniform field at the center of the big loop:

$$B_2 = \frac{\mu_0 I_2}{2r_2}, \quad \text{along the big loop's normal.}$$

Magnetic moment of the small loop: $m_1 = I_1 \pi r_1^2$.
Torque:

$$\tau = \mathbf{m}_1 \times \mathbf{B}_2, \qquad |\tau| = m_1 B_2 \sin\theta.$$

$$\boxed{|\tau| = \frac{\pi \mu_0 I_1 I_2 r_1^2 \sin\theta}{2r_2}.}$$

Direction: perpendicular to the plane containing the two normals, acting to rotate the small loop so its normal aligns with the big loop's field (toward $\theta = 0$).

Summary of Main Results

This set covered every major topic in magnetostatics: the Lorentz force (5-1); Ampère's law in symmetric configurations (5-2, 5-6, 5-10, 5-12); Biot–Savart for finite geometries (5-3, 5-4, 5-5, 5-7, 5-8); the vector potential (5-9, 5-10); magnetization and bound currents (5-11, 5-12, 5-13); boundary conditions (5-14); inductance (5-15, 5-16, 5-17); and forces and torques on current-carrying circuits (5-18 through 5-22).

The recurring methodological themes are: use symmetry to pick the right tool — Ampère's law when you have a symmetric current distribution, Biot–Savart when you don't; exploit the vector potential when the flux through an awkwardly shaped loop would otherwise require a difficult surface integral, since $\Phi = \oint \mathbf{A} \cdot d\mathbf{l}$ is often much cleaner; and remember that $\tau = \mathbf{m} \times \mathbf{B}$ is the shortcut for torque on any small current loop in a field that's roughly uniform over its extent.